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(21216) Roll No. .....

BCA - III Sem.

18013

B. C. A. Examination, Dec. 2016

Computer Architecture and Assembly Language

(BCA-303)

(New Course)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

Section-A

(Very Short Answer Questions)

Answer all the *five* questions. Each question carries equal marks. Very short answer is required not exceeding 75 words. 3×5=15

1. Give important characteristics of RISC architecture.

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2. What is the advantage of using Booth algorithm?

3. Differentiate between micro-instruction and microprogram.

4. What is the difference between a direct and an indirect address instructions?

5. Convert the following into reverse polish notation :

   A * B + C.

Section-B

(Short Answer Questions)

Answer any *two* questions out of the following three questions. Each question carries equal marks. Short answer is required not exceeding 200 words. 7½×2=15

6. Discuss Basic Computer Organization. How is it different from computer architecture?

7. What do you understand by pipeline processing? Explain an arithmetic pipeline that adds two normalized floating numbers.

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8. Write a program in assembly to ADD two numbers stored at memory location 2001H and 2002H. Result will be display at 2004H.

Or

What do you understand by Instruction Cycle? What are the different phases of instruction cycle?

Section-C

(Detailed Answer Questions)

Answer any *three* questions out of the following five questions. Each question carries equal marks. Answer is required in detail. 15×3=45

9. What do you understand by Parallel Processing? Discuss in detail how it helps in reducing the running time. Which kind of tasks can be parallelized?

10. What do you mean by Addressing Models? Explain the following addressing modes :

(a) Immediate (b) Implied.

(c) Direct (d) Indirect.

(e) Relative (f) Base Register.

(g) Indexed.

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11. Describe Direct Memory Access (DMA). Explain its functioning of DMA transfer with the help of diagram.

12. Differentiate between :

(a) General purpose and Special Purpose Register

(b) RISC and CISC instruction set

(c) Hardwired and micro-programmed control

(d) Vector processing and Array processing

(e) Interrupt and subroutine

13. What is Priority Interrupt? Explain Polling and Daisy Chaining Priority.

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18013

B.C.A. - III Sem.

B.C.A. Examination, Dec. – 2017

COMPUTER ARCHITECTURE AND ASSEMBLY

Language (BCA-303) (New)

Time : Three Hours] **(Maximum Marks : 75**

Note : Attempt all sections as per instructions.

Section-A

(Very Short Answer Questions)

Note : Attempt all five questions. Each question carries equal marks. Very short answer is required not exceeding 75 words. 5x3=15

1. What is the use of register? Name any two registers.

2. What do you mean by cycle stealing?

3. Differentiate between hardwired and micro-programmed controls.

4. Explain in brief, how subroutine call is different from a software interrupt?

5. Discuss the stack organization for processor.

Section-B

(Short Answer Questions)

Note : Attempt any two questions. Each question carries equal marks. Short Answers is required not exceeding 200 words. 2x7.5

1. What are the steps required for a pipelined processor to process the instruction? Draw a pipeline for (A+B*D).

2. Write a program in assembly to subtract number stored at memory location 2001H from the number stored at 2002H. Display the result at 2004H.

OR

What is Booth algorithm? Explain its advantages.

3. What is asynchronous data transfer? How! handshaking removes the disadvantage of strobe method?

Section-C

(Detailed Answer Questions)

Note : Attempt any **three** questions. Each question carries equal marks. Answer is required in detail.

3×15=45

1. Draw and explain the architecture of 8085 Microprocessor along with all its registers and instruction set.

2. What do you mean by addressing modes? Explain various addressing modes with the help of examples.

3. What do you mean by Input-output Processor (IOP)? Explain with the help of block diagram

4.What do you understand by Instruction Cycle? What are the different phases of instruction cycle?

5. Write short note on:

(a) RISC/CISC (b) Vector Processor (c) Array Processor (d) Parallel Processor (e) DMA

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(21218) **Roll No. .....**

BCA-III Sem.

18013

B. C. A. Examination, Dec. 2018

Computer Architecture & Assembly Language

(BCA-303)

(New Course)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

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Section-A

(Very Short Answer Questions)

Attempt all the *five* questions. Each question carries 3 marks. Very short answer is required.

3×5=15

1. What are the three major phases through which the control unit go through an instruction cycle ?

2. Write a note on computer registers.

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3. What do you understand by interleaved D. M. A.?

4. What is asynchronous data transfer ?

5. Distinguish between fixed point and floating point representation.

Section-B

(Short Answer Questions)

Attempt any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required. 7½×2=15

6. (a) Differentiate between RISC and CISC.
   (b) What is the difference between hardwired control and microprogrammed ?

7. Draw and explain a 4-bit arithmetic circuit which can perform the following :
   (a) Add
   (b) Add with carry
   (c) Subtract with borrow
   (d) Subtract
   (e) Transfer of A
   (f) Increment
   (g) Decrement.

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8. Write an assembly language program to add 'n' number where the numbers are stored in 'n' consecutive locations (NUM, NUM+1.....NUM + n-1) and to store the result in memory location SUM. The number 'n' is stored in memory location N.

Section-C

(Detailed Answer Questions)

Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. (a) Perform the subtraction of the following unsigned decimal number by taking 10's complement of the subtrahend :

7452 – 1243

(b) Perform the subtraction of the following unsigned binary number by taking 2's complement of the subtrahend :

11010 – 1101

(c) What is the use of macros in I/C instruction ?

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10. Draw a block diagram for data transfer from CPU to an interface and then to an I/O device. Determine the procedure for setting and clearing the flag bit.

11. What is a difference between a direct and indirect address instruction ? How many references to memory are needed for each type of instruction to bring an operand into a processor register ?

12. Draw and explain one stage of an ALU with shift capability along with the microoperations performed.

13. Write short notes on any three of the following :
    (a) Arithmetic pipelining
    (b) Instruction set
    (c) Interrupts useful in improving processing efficiency
    (d) Array processor
    (e) Serial communication.

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(21119)

B.C.A.-III Sem.

Printed Pages : 3

Roll No. .....

18013

B.C.A. Examination, November-2019

COMPUTER ARCHITECTURE AND ASSEMBLY LANGUAGE

(BCA-303)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from ***all*** sections as per instructions.

Section–A

(Very Short Answer Questions)

Note : Attempt all ***five*** questions. Each question carries 3 marks. Very short answer is required not exceeding 75 words. 5×3=15

1. What is the role of Assemblers ?

2. What are Macros ?

3. Define the Computer Registers.

4. Write short note on asynchronous data transfer.

5. What is cache memory ? Describe its operations in brief.

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Section–B

(Short Answer Questions)

Answer any ***two*** questions out of the following three questions. Each question carries 7½ marks. Short answer is required not exceeding 200 words.

2×7½=15

6. Define interrupt. Explain its various types.

7. Differentiate between direct and indirect addressing with an example.

8. Explain subroutine in assembly language.

Section–C

(Detailed Answer Questions)

Attempt any ***three*** questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 3×15=45

9. What is Booth algorithm ? Explain it in detail. Multiply 24 and –7 using Booth algorithm.

10. Write short note on following :

    (i) Synchronous Data Transfer

    (ii) Serial Communication

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(iii) Index Register

11. Describe the concept of DMA used in computer organisation.

12. (a) A CISC chip is a complex instruction set computing chip the alternative to RISC chips. How do they differ ?

(b) Define Instruction Cycle. Explain each phase.

13. Explain the difference between hardwired control and microprogrammed control. Is it possible to have a hardwired control associated with a control memory.

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(Printed Pages 4)

(20221)

Roll No.

BCA-III Sem.

18013

B.C.A. Examination, Dec. - 2020

COMPUTER ARCHITECTURE AND ASSEMBLY LANGUAGE

(BCA-303)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** sections as per instructions.

Section- A

(Very Short Answer Questions)

Note : Answer **all** the **five** questions. Each question carries equal marks. Very short answer is required not exceeding 75 words. 5×3=15

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1. Define and explain Cache memory. 3

2. Differentiate b/w isolated I/O and memory mapped I/O. 3

3. Define & explain Booth's algorithm. 3

4. Write about Flag register in 8085. 3

5. Write an assembly language program to add two nos. 3

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Section - B

(Short Answer Questions)

Note: Attempt any **two** questions. 2×7½=15

6. What are the steps for a simple instruction cycle? Explain Fetch cycle and Indirect cycle using Register transfer language. 7½

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7. What do you understand by DMA. Explain giving a diagram. 7½

8. Explain Programmed I/O with a flow chart. 7½

Section - C

(Detailed Answer Questions)

Note: Attempt any **three** questions. 3×15=45

9. Discuss various logical instructions, Machine Control Instructions and Program Control Instructions in the Assembly Language. 15

10. (a) List five important characteristics of RISC Architecture. 5

(b) Differentiate B/W Hardwired Control Unit Vs Micro-programmed Control Unit. 5

(c) Explain Interrupt Driven I/O in detail. 5

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11. (a) What are various data transfer schemes? Briefly discuss each scheme. 5

(b) Explain the need of different addressing modes by taking suitable examples. 5

(c) Explain the role of register transfer in computer architecture. 5

12. Discuss the following in brief: 15

(a) Program loops in Assembly Language

(b) Operation code

(c) 8 bit Microprocessor

13. Write short notes on: 15

(a) Memory Inlufacing Memory.

(b) Floating point representation

(c) Architecture of 8085.

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A (Printed Pages 4)

(20222) Roll No. .....

BCA-III Sem.

18013 (CV-III)

B.C.A. Examination, Dec.-2021

COMPUTER ARCHITECTURE AND ASSEMBLY LANGUAGE

(BCA-303)

Time : 1½ Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** sections as per instructions.

Section- A

Note : Attempt any **two** questions. Each question carries 7.5 marks.

2×7.5=15

1. Define the Computer Registers.

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2. Differentiate between Micro-instruction and micro program.

3. What is the advantage of using Booth Algorithm?

4. What is cache memory? Describe its operations in brief.

5. Convert the following into reverse polish notation.

   A*B+C

Section - B

Note: Attempt any **one** question. Each question carries 15 marks. 1×15=15

6. Differentiate between direct and indirect addressing with an example.

7. Discuss basic computer organization. How is it different from computer architecture.

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8. Explain subroutine in assembly language.

Section - C

Note: Attempt any **two** questions. Each question carries 22.5 marks. $2 \times 22.5 = 45$

9. What is Booth algorithm? Explain it in detail. Multiply 24 and -7 using Booth algorithm.

10. Describe Direct Memory Access (DMA) Explain its functioning of DMA transfer with the help of diagram.

11. What is Priority Interrupt? Explain polling and Daisy chaining Priority.

12. What do you mean by Input-output processor (IOP)? Explain with the help of block diagram.

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13. Write short note on:

    (a) RISC/CISC

    (b) Array Processor

    (c) Parallel Processor

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A (Printed Pages 3)

(21222) Roll No. .....

B.C.A.-III Sem.

18013

B.C.A. Examination, Dec. 2022

Computer Architecture and Assembly Language

(BCA-303)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** sections as per instructions.

Section-A

(Very Short Answer Questions)

Note : Attempt **all** questions. 5×3=15

1. What is the use of register? Name any two registers.
2. Differentiate between hardwired and micro-programmed controls.
3. Distinguish between fixed point and floating point representation.

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4. What are Macros?
5. What is Cache memory? Describe its operations in brief.

Section-B

Note : Attempt any **two** questions.

2×7½=15

6. Define Interrupt. Explain its various types.
7. What is Booth algorithm? Explain its advantages.
8. Differentiate between RSIC and CISC.

Section-C

Note : Attempt any **three** questions.

3×15=45

9. Describe Direct Memory Access (DMA). Explain its functioning of DMA transfer with the help of diagram.
10. What is Priority Interrupt? Explain Polling and Daisy chaining priority.
11. Differentiate between:

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(a) General purpose and Special purpose register

(b) Vector processing and Array processing

(c) Direct and Indirect address instruction

12. What do you mean by Addressing Modes? Explain various addressing modes with the help of examples.

13. Draw and explain the architecture of 8085 microprocessor along with all its registers and instruction set.

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D (Printed Pages 4)

(21223) Roll No. .....

B.C.A.-III Sem.

18013

B.C.A. Examination, Dec.-2023

COMPUTER ARCHITECTURE AND ASSEMBLY LANGUAGE

(BCA-303)

Time : 3:00 Hours] *[Maximum Marks : 75*

Note : Attempt question from **all** sections as per Instruction.

Section-A

(Very Short Answer Question)

Note : Attempt **all** question. 5×3=15

1. What is Instruction? Explain.
2. Write an assembly language program to add two numbers.

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3. What is Macros.
4. What is the full form of Risc.
5. Write about flag register in 8085.

Section-B

(Short Answer Question)

Note : Attempt any **two** question. 2×7½=15

6. What is serial communication? Explain.
7. What do you understand by DMA. Explain Giving a diagram.
8. Define Interrupt. Explain its various type.

Section-C

(Detail Answer Questions)

Note : Attempt any **three** question. 3×15=45

9. Discuss the following in brief.
   (a) General Register organization
   (b) Index Register
   (c) Input/output processor (IOP)

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10. Write a short notes on:

(a) Memory Influfacing Memory

(b) Architecture of 8085.

(c) Floating point Representation.

11. (a) What is various data transfer schemes? Briefly discuss each scheme.

(b) Explain the need of different addressing mode by taking suitable example.

(c) Explain the role of register transfer in computer Architecture.

12. What is priority Interrupt? Explain polling and chaining periority.

13. (a) List the five important characteristics of Rise Architecture.

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(b) Explain Interrupt driven I/o in detail.

(c) Differentiate between Hardwired control unit VS Microprogrammed control unit.

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